

ECE 2510 Project

Spring 2026

Task 1- Finding a Prime Number (10 points)

Write a subroutine that calculates if an integer (between range 0-9999) is a prime number or not.

- The subroutine should take in a 32 bit unsigned integer, with its value between 0-9999. The return value should be a 32 bit boolean true or false, determining if the value is prime or not.
- The input and output parameters should be passed to the subroutine through the stack
 - Input: 32 bit integer we are testing
 - Output: boolean value to determine whether the value is a prime number
 - 0: not a prime number
 - 1: is a prime number
- Draw your flow chart and the stack frame. Comment your code for readability. The pseudocode is given for you below.
 - Note: it may be better for you to write 2 subroutines, 1 for finding the square root of the value and another for finding if the value is a prime. Below are some guidelines for your subroutine to find a prime number.

```
ALGORITHM IsPrime(n)
  IF n < 2 THEN
    RETURN False
  ENDIF
  FOR i = 2 TO square_root(n)
    quotient = n DIV i
    product = quotient * i
    IF product = n THEN
      RETURN False
    ENDIF
  ENDFOR
  RETURN True
END ALGORITHM
```

Task 2- Finding Primes in an Array (5 points)

Now that you have written a subroutine to identify prime numbers, you can use it to detect all prime numbers in an array of integers. For this task, write a subroutine that finds all the prime numbers in one array and stores them in another array.

- Create a 32 bit integer array called input, with each value in the array being between 0-9999
- Create a second result array to hold the prime numbers found in input
- Your program should loop through the first array and call the prime isPrime subroutine on each element
- When a value is found to be a prime number, store it in the result array
- Draw your stack frame and flowchart for this task as well.

Task 3- 7-Segment Display (10 points)

Now that we have found the primes and stored them in the result array, we need to display them. For this task, write a subroutine that displays all numbers in the result array on the 7- segment display, one number at a time with a one-second delay after each number. Note that you need to use time multiplexing to do this. Your code should display the prime numbers one after the other from the result array and once the last prime number is displayed, the program should repeat and display them again all the time. Your code should be well commented for full credit.

Bonus Tasks:

You can choose to do any or all of the following bonus tasks. The bonus will count towards your total class grade. Comment your code for full credit.

Bonus 1- Port C Interrupts (8 points)

For bonus 1, use interrupts on Port C to take an input from the dipswitch while your task 3 code is already running. In the ISR, your code should read in the value from the dipswitch and signal if it is a prime number or not. If the value is prime, display “YES” on the 7 segment display for 3 seconds, blinking on and off in .5 second intervals before resuming the code from task 3. If the

value is not a prime number, then display “NO” on the 7-segment display for 3 seconds, blinking on and off for .5 second intervals before resuming the code from task 3.

Bonus 2- Making Delays Using Timer Registers (8 points)

Execution loops with NOPs or PUSH and POP are not the only ways to make delays in ARM. The architecture has timer registers meant for this purpose. For the second bonus, write a code that will configure a timer register to create the delays needed for the 7 segment (display as well as time multiplexing).

Project Report (5 points)

The format of the project will be the same as the lab report given throughout the semester.

1. The report should contain the following:
 - a. Description of your project, methods used, and challenges.
 - b. Assembly code – well commented.
 - c. Flowcharts for each task including the bonus tasks.
 - d. Stack diagrams for each subroutine.
 - e. For task 1 and task 2, include a screenshot of the number of cycles used to run the code.
 - f. Provide in detail the contribution of each group member to the project.
2. If you have used any reference material from the internet or book, please cite all references at the end of the report.
3. In addition to your report, submit a zipped copy of your program folder

Submission Rules

1. A Demo of your project to the TA (Required!).
2. Any form of plagiarism will result in dismissal of the project grade and reported to the course instructor.
3. All project reports are individual.